Milestone 3

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Research Question:
What is the minimal set of non-redundant, 24-hour clinical features (vitals, labs, demographics) that are the most impactful predictors of in-hospital ICU mortality? Furthermore, can this core set of features be used to train a reliable triage model that performs well despite the challenges of high class imbalance and data missingness?

Describe the problem
The Clinical Challenge
Accurately predicting a patient's risk of death in the Intensive Care Unit (ICU) is a critical but difficult task. Real-world clinical decision-making is famously complex, and in the high-pressure ICU environment, it is easy to get lost in the plethora of available clinical data. This complexity highlights the urgent need for an interpretable and reliable predictive tool to support patient, patient triage, resource allocation and early prognosis.

The Data Science Challenge
The data science problem is to build a reliable and interpretable predictive model that can estimate this risk using only the data collected within the first 24 hours of an ICU admission. The model must be robust enough to handle the severe data quality challenges of a real-world clinical dataset, most notably extreme data missingness and high class imbalance.

Our Dataset
We are using the ‘training_v2.csv’ file from the WiDS Datathon 2020 repository (Fig 1). It contains 91,713 ICU encounters, aggregated from over 200 hospitals, with 185 features used to predict the target variable, ‘hospital_death’ (i.e. in-hospital mortality).

Brief Data Overview
The data is overwhelmingly numeric (171 float64 columns) with a few categorical (object) features. All features were captured within the first 24 hours of a patient's ICU stay and include demographics, vital signs, laboratory values, and APACHE covariates, which are metrics from the APACHE IVa scoring system.

With our initial exploration, three primary challenges became apparent:

1. Feature Complexity: The dataset is highly complex, with over 180 features and significant multicollinearity, as visualized in the correlation heatmap (Fig. 2a). This finding directs our methodology toward regularization techniques, like Lasso, to perform automatic feature selection.
2. Class Imbalance: The target variable, hospital_death, is highly imbalanced, with a 10.6-to-1 survivor-to-death ratio (Fig. 2b). This imbalance makes accuracy a misleading metric, as a 'naive' model that only predicts 'survive' would achieve over 91% accuracy. This finding necessitates using robust evaluation metrics (e.g., AUROC) and modeling strategies (e.g., class weighting).
3. Systematic Missingness: Missing data is a core problem. A clear "two-camp" pattern emerged: 175 columns have missing values, but 74 of those are missing over 50% of their data. This systematic missingness informs our pipeline's strategy of dropping low-quality features to focus on a high-quality subset. Fortunately, none of the target variables is missing.

We will discuss our specific solutions for these challenges at length in the Data Preprocessing Pipeline section below.

Data Preprocessing Pipeline

First, we filtered out features with excessive missingness (>25%) to focus on higher-quality data, reducing the feature space from 182 to 112 columns. For the remaining missing values, we implemented a sophisticated imputation strategy that diagnoses missingness patterns using logistic regression models to predict missing value indicators. For features with high AUROC scores (>0.75) indicating Missing At Random receive KNN imputation to preserve complex relationships, while features with low AUROC scores indicating Missing Completely at Random receive simple median/mode imputation. This methodology accepts the risk of Missing Not at Random error because of the high quality of the data source (medical professionals) as well as the fundamentally paradoxical nature of this type of missingness.

After imputation, we apply Lasso regularization across a large range of lambda values to automatically select the most predictive features while addressing multicollinearity, ultimately identifying the top ~20 features that provide optimal predictive power. This set of features is listed in Fig 3.

This cutoff of ~20 is a bit arbitrary, and looking at the clustered heatmap in Fig 9 we saw some evidence that multicollinearity remains in this set of features. So we used VIF scores to prune further. This identified 3 redundant blood pressure values that we will be removing. See Fig 8 for more details on this.

This pipeline transforms a complex, incomplete dataset into a clean, focused feature set suitable for predictive modelling.

Analysis of Cleaned Dataset

Distribution Plots

We first plotted the distribution (Fig. 4) of the mortality rates across different hospitals.This wide inter-hospital variance is a critical finding. It strongly suggests that patient outcomes are dependent on which hospital they are in, not just their individual clinical features. This finding directly impacts our project direction, as it justifies exploring a hierarchical (or multilevel) logistic regression.

After imputing missing values, we also wanted to explore whether imputation affects the distribution of our predictor variables. Fig. 5 shows the typical binary distribution of our predictors. The majority category dominates the data, with our simply imputed values negligibly changing the balance of these categories. We believe it best to simply impute, with `SimpleImputer`, the majority category, since it is less impactful to add a few entries to the majority category than to add the same number (but proportionally larger) of entries to the minority category. Additionally, many of these binary minorities are purportedly highly predictive, and we have decided against diluting the effect of, say, the presence of leukemia in the model with falsely positive imputations.

Fig. 6 shows the two typical patterns for the distributions of our numeric predictors. The higher missingness features imputed with SimpleImputer() are roughly normal, but with large imputed spikes at their medians. This may slightly dampen the inferred coefficient for, say, minimum temperature, as a consequence for the feature's missingness. Though unclear now, we may be able to ameliorate this side-effect of imputation by introducing random noise into our simple imputation process to smooth out the distribution. Also, since most of our numeric features are normal, we may want to consider transforming a feature like mean blood pressure into a deviance measure from the norm (e.g. square distance from the median, creating features for lower/higher than healthy to interact with).

Outlier Analysis

Fig. 7 shows box plots below typical of the distributions of our numeric features. Though each box plot shows numerous outliers, these outliers resemble extended tails on the distribution. Combined with the fact that these are medical measurements with low probability of misinput/mismeasurement, as well as the human naturally bounded nature of biological statistics, we do not see distortingly deviant data points. If anything, these outliers are valuable to us as high-variance data, helping establish the coefficients themselves, rather than damaging our model.

Basic Model

For our final 18 features, we fit a multivariate Logistic Regression model as a baseline. We used a train test split of 4:1. The results highlight the classic trade-off of an imbalanced dataset: while the ROC AUC (0.859) is high, indicating strong discriminative power, the classification performance is poor. The model achieved modest Precision (0.635) but extremely low Recall (0.217), meaning it failed to identify ~75% of all patient deaths. This clinically unusable baseline confirms our next steps must focus on boosting recall, for example, applying class weighting or SMOTE resampling.

Noteworthy Findings

1. The 10.6:1 survivor-to-death ratio makes 91% naive accuracy useless, forcing our use of AUC and F1-score to evaluate model performance.
2. High multicollinearity and a "two-camp" missingness pattern (74 columns >50% missing) justified using Lasso and dropping low-quality features.
3. Mortality rates vary significantly by hospital (Fig. 4), justifying the exploration of a hierarchical model to account for group-level effects.
4. Our baseline model proved our 18 features are highly predictive (AUROC 0.859) but clinically useless (Recall 0.217), proving our next steps must focus on boosting recall.